# Unit 4 Project

**Question 1:**

1a) 1.165%

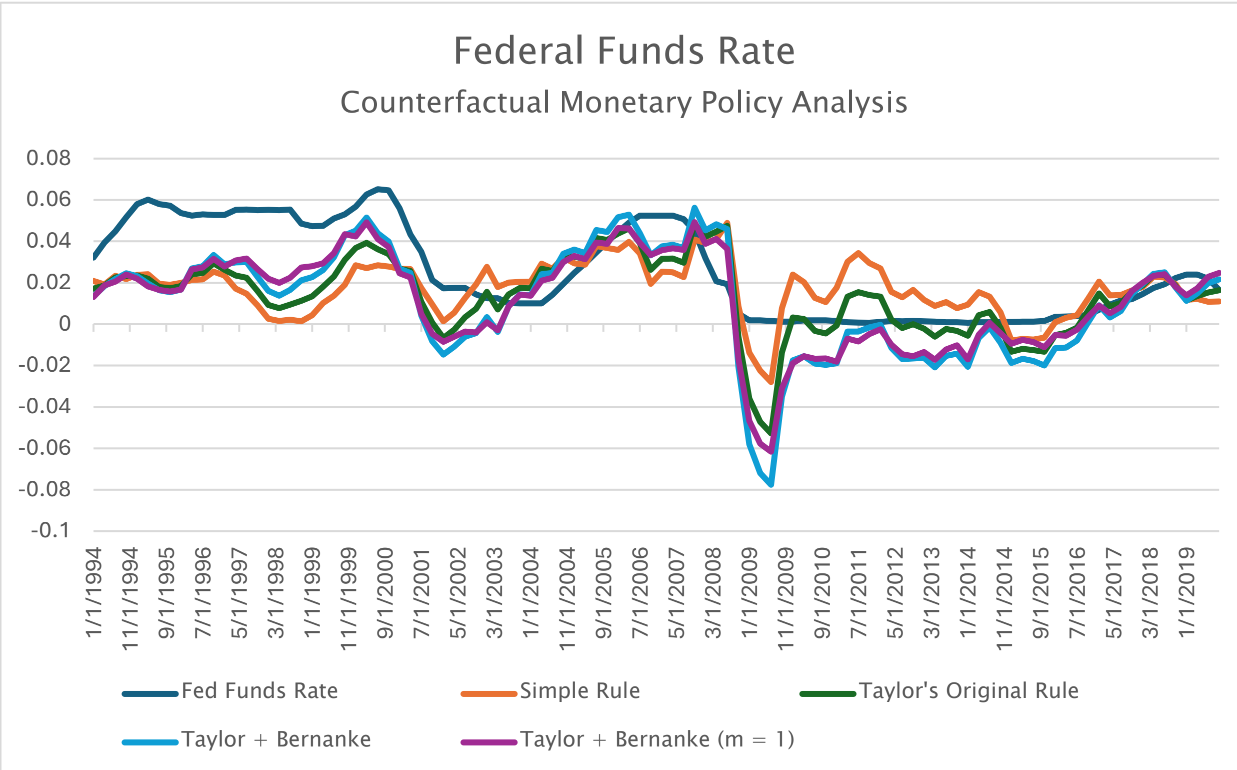
1b) 2.009%

1c) 2.854%

1d) 2.632%

1e) The 2025Q2 Effective Federal Funds Rate was 4.33%. This suggests that none of the four rules from parts (a)-(d) are accurately capturing the Fed's current operating procedure. An upward adjustment of sensitivity to the output gap from 1 to 2, gets very close to the observed EFFR. While a doubling of sensitivity to the output gap is unlikely, it does appear that the current Fed operating procedure prioritizes managing risk and expectations. Furthermore, the unique post-COVID economic environment which led to high levels of inflation has only very recently begun to return to more typical levels. The Fed may be keeping rates higher to ensure that inflation continues to decline and that prices stabilize sustainably.

**Question 2:**




2a)

| Correlations | |
|---:|---|
| Simple Rule v FFR | 0.344 |
| Taylor Original v FFR | 0.639 |
| Taylor + Bernanke v FFR | 0.735 |
| Taylor + Bernanke (m=1) v FFR | 0.771 |

The rule that appears to best describe the historical evolution of the federal funds rate and demonstrates the highest correlation with observations, is the Taylor rule with $\bar{n} = 1$ and $\bar{m} = 1$.

2b)

The estimated coefficients are $\bar{m} = .551 \quad and \quad \bar{n} = .863$. This suggests that the fed is historically more sensitive to the output gap rather than inflation alone and is consistent with the rule that appeared most accurate out of the available options as it is the only rule that shifted relative weight away from inflation sensitivity and towards the output gap.

2c)

The findings outlined in this question, specifically that monetary policy rules which shift relative weight towards output gap sensitivity better represent historical federal funds rates, aligns with the findings in question 1e. While current levels of inflation and the positive output gap are beginning to trend downward, the economy is still normalizing from higher-than-normal inflation and productivity levels which are above potential. As indicated in the regression analysis, the fed has historically been more sensitive to output gap and, with the current gap of 1.69%, this sensitivity explains why rates may be higher than expected based on the value of PCE inflation alone.

3a)

Though I determined that none of the monetary policy rule equations represented the Fed's current operating procedure, the one that came closest was the equation from 1c where $\bar{m} = 1.5 \; and \; \bar{n} = 1$.

$$i_t = .005 + .015(\pi_t - .02) + 1\tilde{Y}_t$$

3b)

$$2025\text{:} \;\; i_t = .005 + .015(.021 - .02) + 1(.01749) = 2.251\%$$

$$2026\text{:} \;\; i_t = .005 + .015(.02 - .02) + 1(.01487) = 1.987\%$$

$$2027\text{:} \;\; i_t = .005 + .015(.02 - .02) + 1(.01295) = 1.795\%$$

**Are the Fed's projections consistent with the monetary policy rule?**

While the Fed's projections on the FFR also trend downward from 2025 to 2027, the Fed's projections are all roughly 1% higher than the monetary policy rule equation suggests. This result is similar to the results observed in question 1 where the monetary policy equations all underestimate the FFR when compared to the observed value.

**How do you think the Fed's internal projections of potential real GDP ($\bar{Y}_t$) might differ from the CBO's?**

Given the model's underestimation of the FFR values when compared to the Fed's projection in the SEP table, it is likely that the Fed is projecting potential GDP to be lower than the projections provided by the CBO. An output gap that is roughly 1% higher would better align the monetary policy equation results with the FED's projected FFR values. This implies that the Fed views the economy as operating farther above potential than the CBO, leading to higher rates.